MIL-M-28787/163A(EC)
10 November 1983
SUPERSEDING
MIL-M-28787/163(EC)
17 March 1978

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, INTERFACE, DIGITAL, KEY CODE EKC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface driver module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.792 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EKC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/163-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

* 3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage of 20 minimum to 35 V dc maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

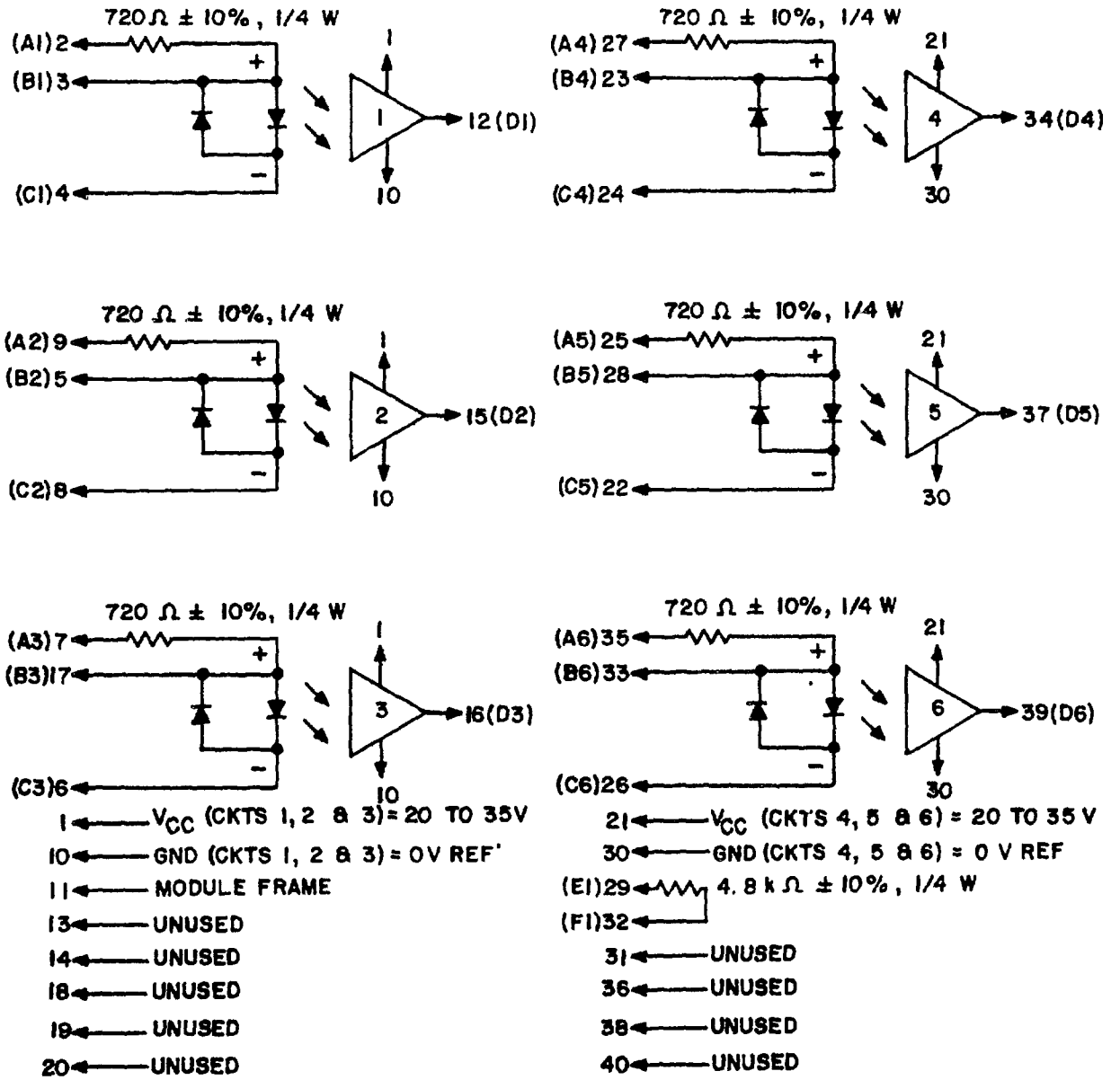
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. A1 - A6 = Current limited input.
2. B1 - B6 = Direct LED input.
3. C1 - C6 = A and B signal return.
4. D1 - D6 = Circuit output.
5. E/F = Series resistor.

FIGURE 1. EKC module function diagram, interface driver.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A11	Isolation	III	X	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B6	Common mode voltage (measuring output leakage current) check	IV	X	0	4	0	4	0	4	0	4	µA
B7-B12	Common mode voltage (measuring output leakage current) check	IV	X	-1	4	-1	4	-1	4	-1	4	µA
C1-C6	Current limiting resistance	V	X	678	762	664	776	667	766	649	783	Ω
C7	Current limiting resistance	V	X	4,524	5,070	4,428	5,166	4,450	5,100	4,308	5,215	Ω
C8	Frame continuity	V	X	0	1	0	1	0	1	0	1	Ω
D1-D6	Input protection diode voltage (I_F = 3 mA)	VI	X	500	710	500	710	430	780	380	780	mV
D7-D12	Output protection diode voltage (I_F = 3 mA)	VI	X	500	750	500	750	430	820	430	820	mV
D13-D18	Output protection diode voltage (I_F = 200 mA)	VI	Q	0.6	1.2	0.6	1.2	0.6	1.2	0.6	1.2	V
E1-E6	Output leakage current	VII	X	0	4	0	4	0	4	0	4	µA
F1-F6	Output saturation voltage	VIII	X	0	600	0	600	0	600	0	600	mV
F7-F12	Output saturation voltage	VIII	Q	0	500	0	500	0	500	0	500	mV
G1-G6	Output saturation voltage (I_{OH} = 170 mA)	IX	Q	0	1	0	1	0	1	0	1	V
H1,H2	Power supply current, outputs on	X	X	17	25	17	25	17	25	17	25	mA
H3,H4	Power supply current, outputs off	X	X	2.2	3.0	2.15	3.10	2.2	3.0	2.15	3.10	mA
J1-J6	Input diode forward voltage (I_F = 0.25 mA)	XI	S	0.8	1.5	0.8	1.5	0.70	1.58	0.70	1.58	V
J7-J12	Input diode forward voltage (I_F = 10 mA)	XI	S	0.8	1.5	0.8	1.5	0.70	1.58	0.70	1.58	V
K1-K6	Transition time (t_{TLH})	XII	S	0.01	10.00	0.01	10.00	0.01	10.00	0.01	10.00	µs
L1-L6	Transition time (t_{THL})	XII	S	0.1	100.0	0.1	100.0	0.1	100.0	0.1	100.0	µs

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M6	Propagation delay time (t_{PLH})	XII	S	0.1	40.0	0.1	40.0	0.1	40.0	0.1	40.0	µs
N1-N6	Propagation delay time (t_{PHL})	XII	S	10	1,000	10	1,000	10	1,000	10	1,000	µs
P1,P2	Parallel output saturation voltage	XIII	S	0	600	0	600	0	600	0	600	mV
Q1-Q6	Output crosstalk, low level	XIV	Q	0	2	0	2	0	2	0	2	V
R1-R6	Output crosstalk, high level	XIV	Q	0	250	0	250	0	250	0	250	mV
S	Power supply short term overload	XV	S	Must pass all functional tests - see 4.4.2 for order of testing.								

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through S shall be performed in any order with the following exceptions. If test conditions S, A, or B are part of a sequence of tests, the order of testing shall be as follows: S, A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through S in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through R shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/163-1 supersedes part number M28787/163. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage	20 V dc minimum, 35 V dc maximum
V_{CC} input current (power dissipation) at V_{CC} = 35 V and R_L = ∞ 1/	
V_{CC} applied to pin 1 and pin 21	
I_{IH} > 3 mA (outputs on)	50 mA maximum (1.75 W)
I_{IL} < 0.25 mA (outputs off)	6.2 mA maximum (0.217 W)
V_{CC} applied to pin 1 or pin 21	
I_{IH} > 3 mA (outputs on)	25 mA maximum (0.875 W)
I_{IL} < 0.25 mA (outputs off)	3.1 mA maximum (0.109 W)

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Input power dissipation (typical per circuit)	
A/C (V_{IH} = 5 V)	27 mW
B/C (V_F LED = 1.1 V, I_F = 5.4 mA)	6 mW
E/F and A/C (V_{IH} = 28 V)	136 mW
Differential low-level input voltage (V_{IL}) for $I_{IL} \leq 0.25$ mA	
A/C (5 V input)	0.86 V maximum
B/C (direct input)	0.7 V maximum
E/F and A/C (28 V input)	1.9 V maximum
Differential low-level input current (I_{IL})	
All inputs	0.25 mA maximum
Input diode forward voltage	
0.25 mA $\leq I_{IL} \leq$ 10 mA	0.7 V minimum, 1.58 V maximum
Output diode forward voltage	0.6 V minimum, 1.2 V maximum
Differential high-level input voltage (V_{IH}) for 3 mA $\leq I_{IL} \leq$ 10 mA	
A/C (5 V input)	4 V minimum, 7.3 V maximum
E/F and A/C (28 V input)	19.5 V minimum, 35 V maximum
Differential high-level input current (I_{IH})	
All inputs	3 mA minimum, 10 mA maximum
Output saturation voltage (paralleled circuits 1, 2, 3, or 4, 5, 6, R_L = 100Ω) (see 6.5.4)	
V_{CC} = 35 V, I_{IH} = 3 mA	0.6 V maximum
Output saturation voltage ($V_{CE\ (sat)}$)	
I_{IH} = 3 mA, I_{OH} = 115 mA, V_{CC} = 35 V	0.6 V maximum
I_{IH} = 3 mA, I_{OH} = 65 mA, V_{CC} = 20 V	0.5 V maximum
V_{IH} = 3.2 V, I_{OH} = 170 mA, V_{CC} = 28 V	1 V maximum
Output drive current maximum capability 2/	
I_{OH} (I_{IH} = 3 mA, V_{CC} = 35 V)	115 mA (300 Ω load)
I_{OH} (I_{IH} = 3 mA, V_{CC} = 20 V)	65 mA (300 Ω load)
I_{OH} (V_{IH} = 3.2 V, V_{CC} = 28 V)	170 mA (158 Ω load)
Current limiting resistance (pin programmable)	
A/B	720 Ω ±10%, 1/4 W
E/F	4.8 kΩ ±10%, 1/4 W
Isolation resistance (between input network and remaining pins) at 100 V	1 MΩ minimum
Output leakage current I_{IL} = 0 mA, V_{CC} = 35 V, V_{ICR} = 100 V	4 µA maximum
Output leakage current I_{IL} = 0.25 mA, V_{CC} = 35 V	4 µA maximum
Switching characteristics (R_L = 10 kΩ	
t_{PLH}	0.1 µs minimum, 40 µs maximum
t_{TLH}	0.01 µs minimum, 10 µs maximum
t_{PHL}	10 µs minimum, 1,000 µs maximum
t_{THL}	0.1 µs minimum, 100 µs maximum

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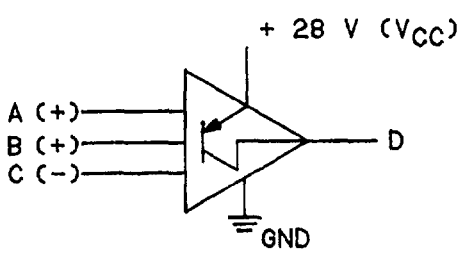
1/ When signals are applied to the module, power from the source is dissipated in the resistor and diode input network. Power ratings of individual components at pins 2, 9, 7, 27, 25, 35, 29, 32, or total package power of 2.5 watts at 60°C (including V_{CC} dissipation) should not be exceeded.
2/ Saturation condition represents two parallel number 7341 lamps (65 mA ±30% at 28 V); see life-test circuit for transient load.

* 6.5 Module function. This module contains six 5 V/28 V isolator circuits, each circuit consisting of a light emitting diode (LED) optically coupled to a photosensitive amplifier with an open collector transistor output capable of sourcing current from a positive supply voltage.

6.5.1 Input to output relationship. Each isolator circuit has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Input	Output
A, B, C	D
I_{IH} or V_{IH}	V_{OH} 1/
I_{IL} or V_{IL}	$I_{LEAKAGE}$



1/ $V_{OH} = V_{CC} - V_{CE(sat)}$.

6.5.2 Maximum input conditions. Input B is not current limited. Input voltages in excess of V_{IF} minimum (0.7 V) or input currents in excess of I_{IH} maximum (10 mA) will cause degradation of the LED. In ac operation, or with pulsed sources, the LED will tolerate peak currents higher than the dc rating if pulse duration is less than 1 millisecond and the duty cycle is less than 50 percent. Under these conditions, the dc rating applies to the average, rather than the peak current. For example, at 1 ms duration, 50 percent duty cycle, LED forward current could be 20 mA_{pk}.

6.5.3 High-voltage input applications. For input voltages between 19.5 V and 35 V, the 4.8 kΩ resistor between pins 29 and 32 must be used in series with one of the 720 Ω resistors available at pins 2, 7, 9, 25, 27, 35. There is one 4.8 kΩ resistor per module.

6.5.4 Parallel outputs. Paralleling of circuits is permissible. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within the groups specified below. The output of this combination will drive a 150 Ω load when two circuits are paralleled and a 100 Ω load when three circuits are paralleled. The circuit groups which can be paralleled are circuits 1, 2, and 3 and circuits 4, 5, and 6.

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6.5.4.1 Wired "OR" function. Wire OR'ing between outputs is limited to circuits with common power supply (V_{CC}) connections. The circuits that can be unconditionally OR'd in the same module are circuits 1, 2, and 3 and circuits 4, 5, and 6.
CAUTION: Damage to the module is possible if power (V_{CC}) is interrupted to one circuit wire OR'd and not to others.

6.5.5 Input polarity. Observe polarity of input voltage.

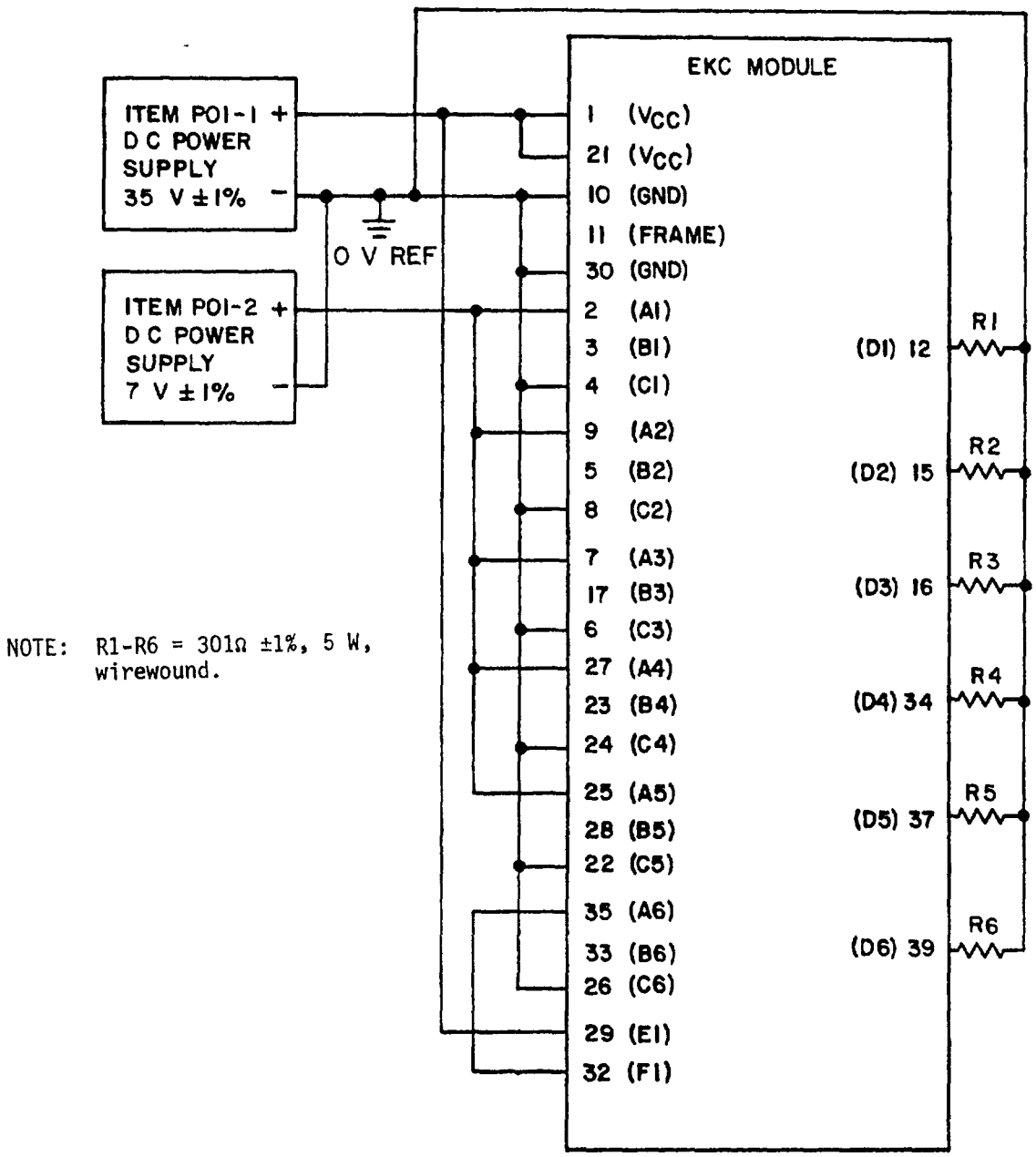
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTE: R1-R6 = 301Ω ±1%, 5 W, wirewound.

FIGURE 2. Operating temperature test circuit.

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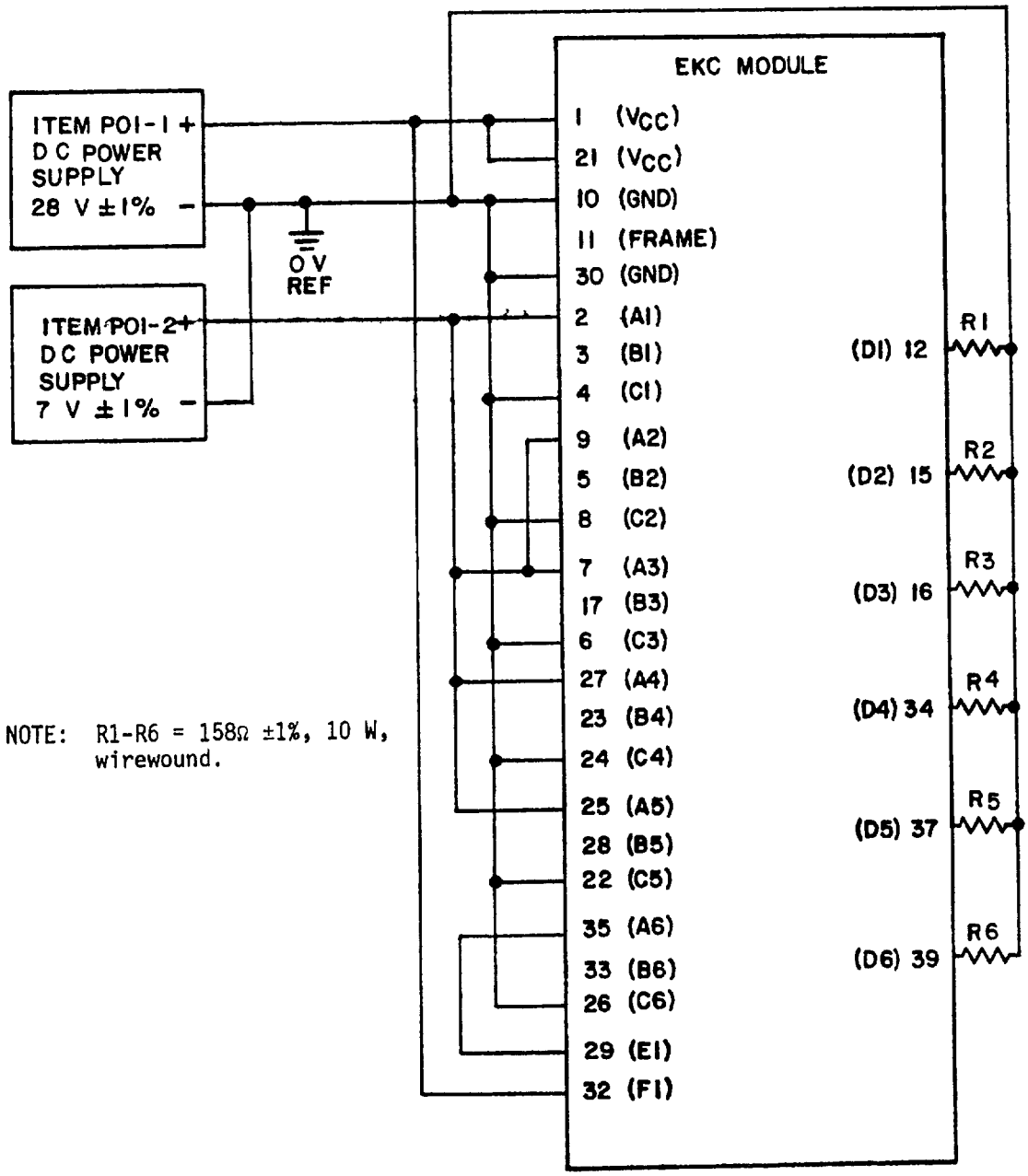


FIGURE 3. Component temperature test circuit.

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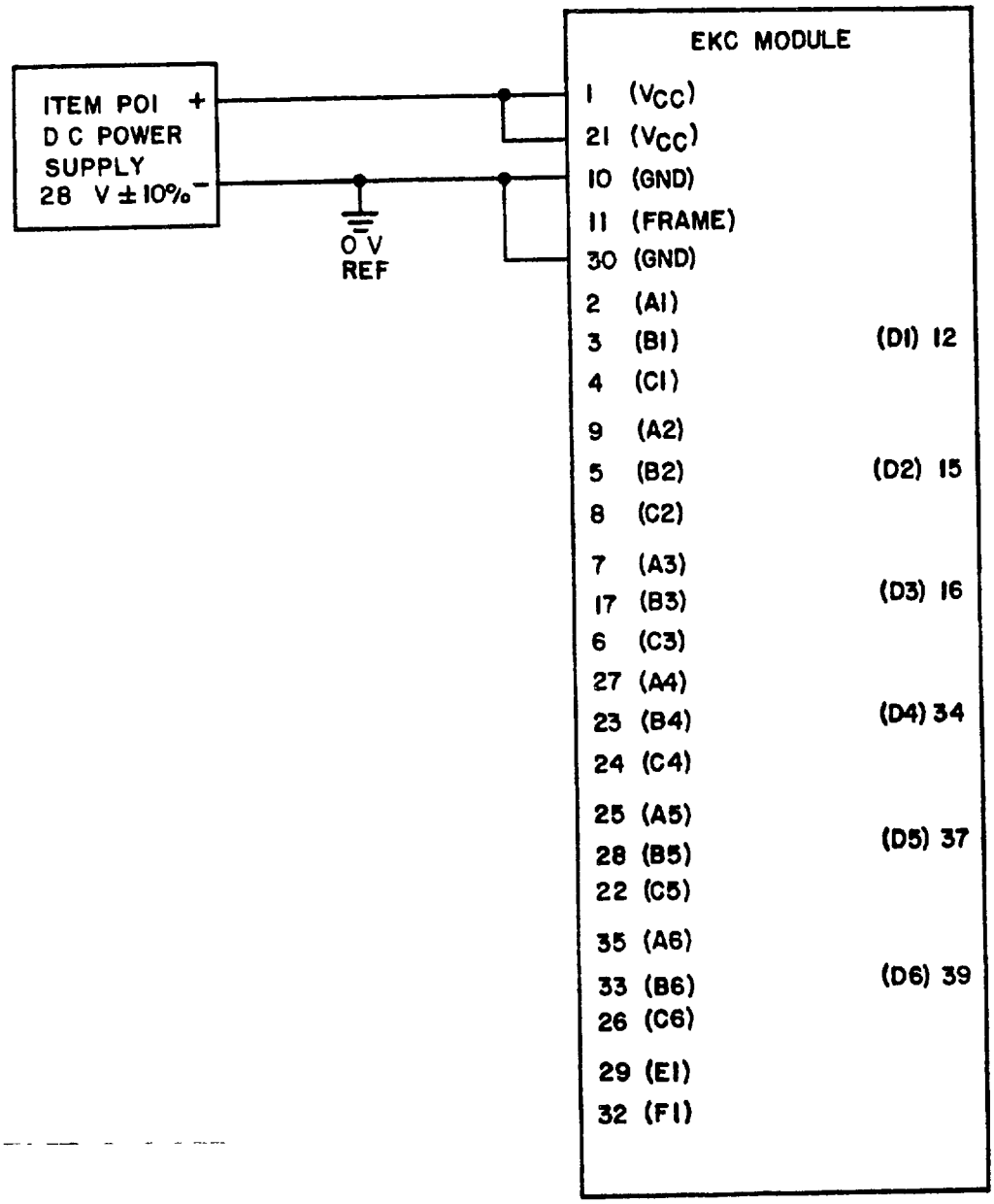
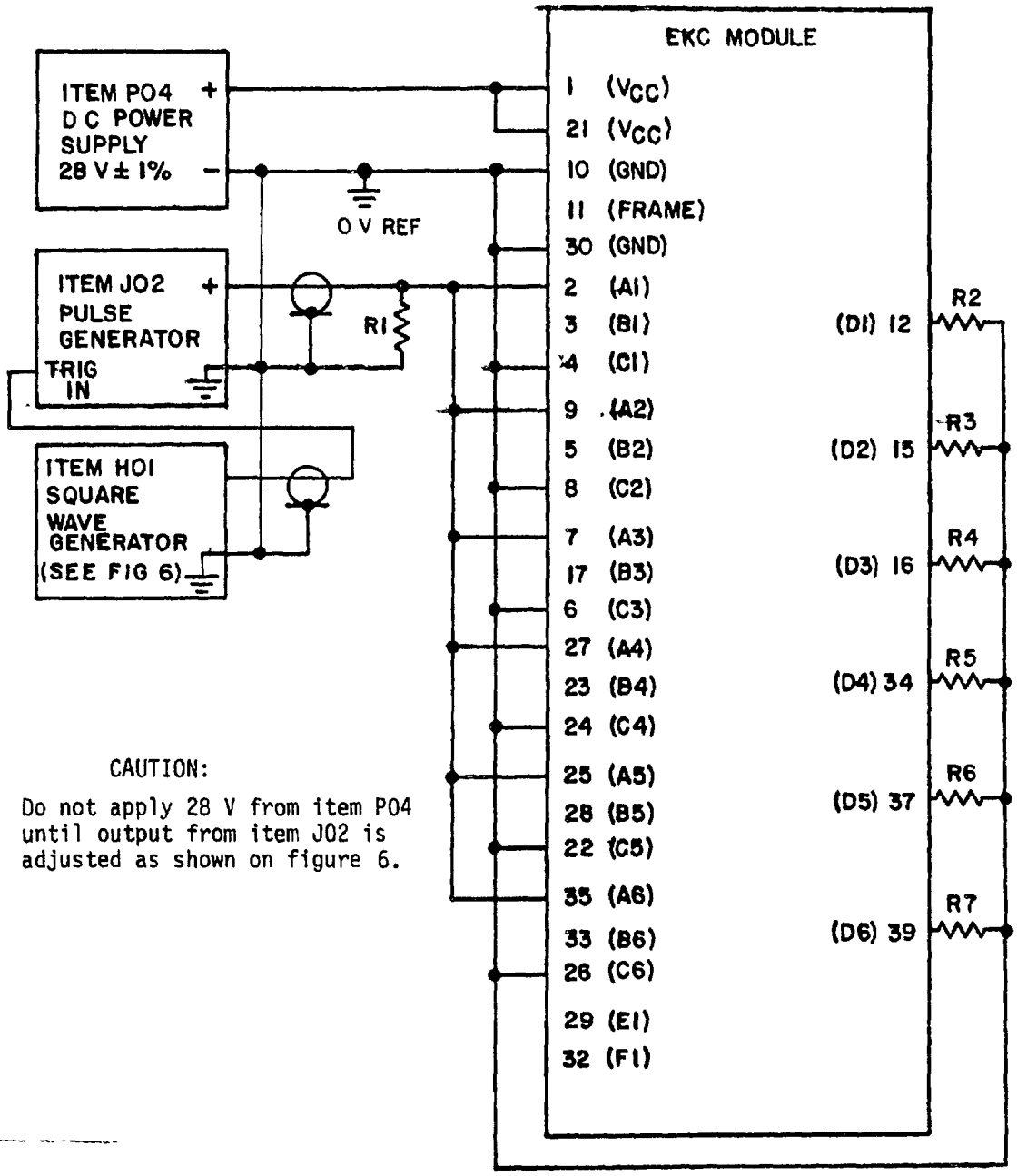


FIGURE 4. Vibration test circuit.

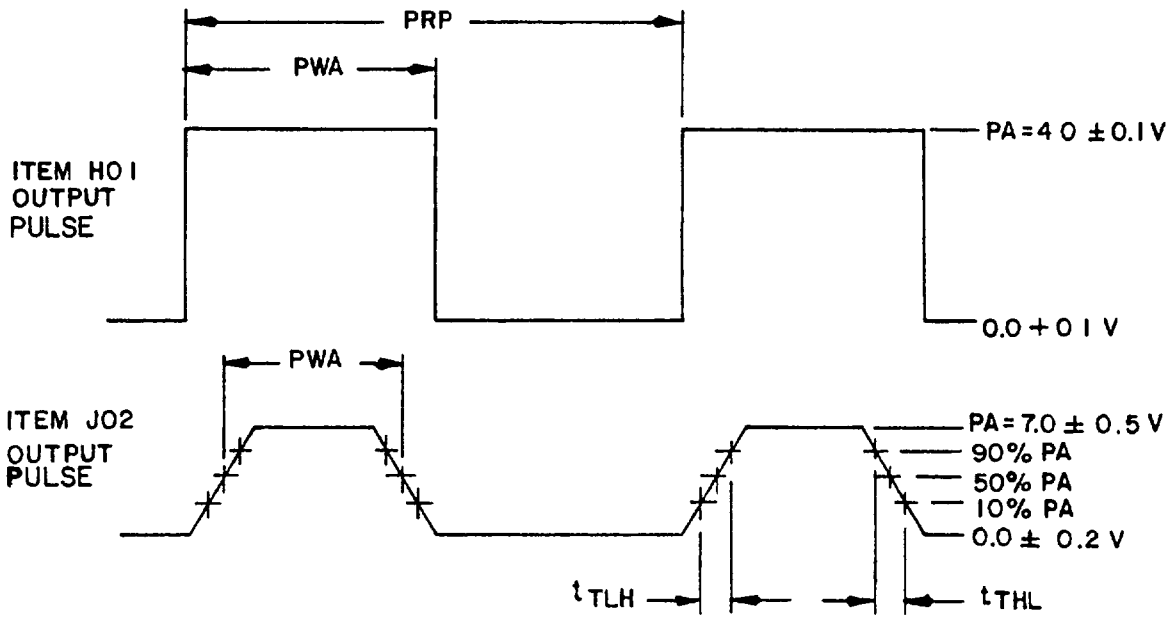
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NOTES:
1. R1 = 51Ω ±5%, 2 W.
2. R2 - R7 = 32.4Ω ±1%, 1/4 W.

* FIGURE 5. Life test circuit.

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NOTES:
1. Item HO1 pulse parameters:
   PWA = 2.5 ±0.5 s.
   PRP = 5 ±1 s.
2. Item JO2 pulse parameters:
   PWA = 15 $^{+0.0}_{-0.5}$ ms.
   $t_{TLH} = t_{THL} = 50 \pm 5$ ns.

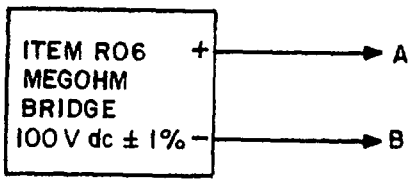
FIGURE 6. Life test circuit waveforms.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, all pins to pin 11	All pins except pin 11	11	10	∞	MΩ	7
A2	Isolation, used to unused pins	1,2,3,4,5,6,7,8,9,10,11,12,15,16,17,21,22,23,24,25,26,27,28,29,30,32,33,34,35,37,39	13,14,18,19,20,31,36,38,40	"	"	"	"
A3	Circuit 1 Isolation, input network and unused pins	2,3,4,13,14,18,19	All other remaining pins (tied togetner)	"	"	"	"
A4	Circuit 2 "	5,8,9,13,14,20,31,36	"	"	"	"	"
A5	Circuit 3 "	6,7,17,13,18,20,36,38	"	"	"	"	"
A6	Circuit 4 "	20,23,24,27	"	"	"	"	"
A7	Circuit 5 Isolation, input network	22,25,28	"	"	"	"	"
A8	Circuit 6 Isolation, input network	26,33,35	"	"	"	"	"
A9	Isolation, power supply	1	21	"	"	"	"
A10	Isolation, ground return	10	30	"	"	"	"
A11	Isolation, current limiting resistance (28 V)	29,32	All other remaining pins (tied together)	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

EKC MODULE		
1	(V_{CC})	
21	(V_{CC})	
10	(GND)	
11	(FRAME)	
30	(GND)	
2	(A1)	
3	(B1)	(D1) 12
4	(C1)	
9	(A2)	
5	(B2)	(D2) 15
8	(C2)	
7	(A3)	
17	(B3)	(D3) 16
6	(C3)	
27	(A4)	
23	(B4)	(D4) 34
24	(C4)	
25	(A5)	
28	(B5)	(D5) 37
22	(C5)	
35	(A6)	
33	(B6)	(D6) 39
26	(C6)	
29	(E1)	
32	(F1)	

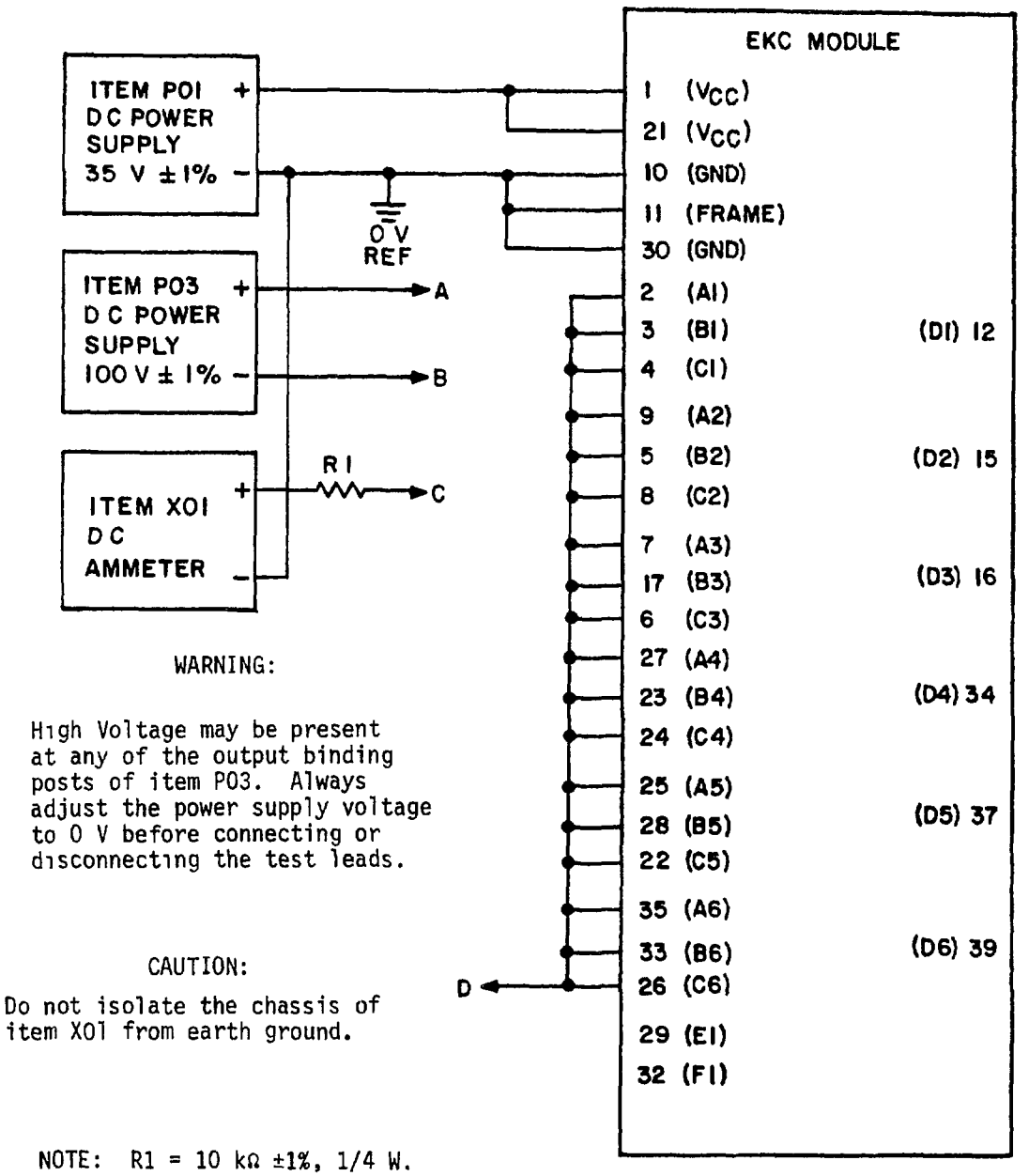
FIGURE 7. Isolation test circuit for table III.

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TABLE IV. Common mode voltage (measuring output leakage currert) check test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B1	D1	Common mode voltage (measuring output leakage current) check	D	30	12	0	4	µA	8
B2	D2	"	"	"	15	"	"	"	"
B3	D3	"	"	"	16	"	"	"	"
B4	D4	"	"	"	34	"	"	"	"
B5	D5	"	"	"	37	"	"	"	"
B6	D6	"	"	"	39	"	"	"	"
B7	D1	"	30	D	12	-1	"	"	"
B8	D2	"	"	"	15	"	"	"	"
B9	D3	"	"	"	16	"	"	"	"
B10	D4	"	"	"	34	"	"	"	"
B11	D5	"	"	"	37	"	"	"	"
B12	D6	"	"	"	39	"	"	"	"

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WARNING:

High Voltage may be present at any of the output binding posts of item P03. Always adjust the power supply voltage to 0 V before connecting or disconnecting the test leads.

CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = 10 kΩ ±1%, 1/4 W.

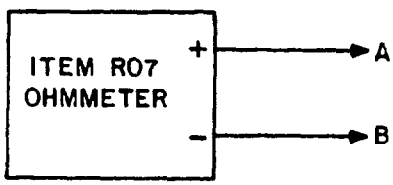
FIGURE 8. Common mode voltage (measuring output leakage current) check test circuit for table IV.

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TABLE V. Current limiting resistance and frame continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A/B 1 Current limiting resistance	2	3	678	762	Ω	9
C2	A/B 2 "	9	5	"	"	"	"
C3	A/B 3 "	7	17	"	"	"	"
C4	A/B 4 "	27	23	"	"	"	"
C5	A/B 5 "	25	28	"	"	"	"
C6	A/B 6 "	35	33	"	"	"	"
C7	E/F 1 "	29	32	4,524	5,070	"	"
C8	Frame continuity	11	Frame	0	1	"	"

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EKC MODULE		
1	(V_{CC})	
21	(V_{CC})	
10	(GND)	
11	(FRAME)	
30	(GND)	
2	(A1)	
3	(B1)	(D1) 12
4	(C1)	
9	(A2)	
5	(B2)	(D2) 15
8	(C2)	
7	(A3)	
17	(B3)	(D3) 16
6	(C3)	
27	(A4)	
23	(B4)	(D4) 34
24	(C4)	
25	(A5)	
28	(B5)	(D5) 37
22	(C5)	
35	(A6)	
33	(B6)	(D6) 39
26	(C6)	
29	(E1)	
32	(F1)	

NOTE: For test condition C8, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

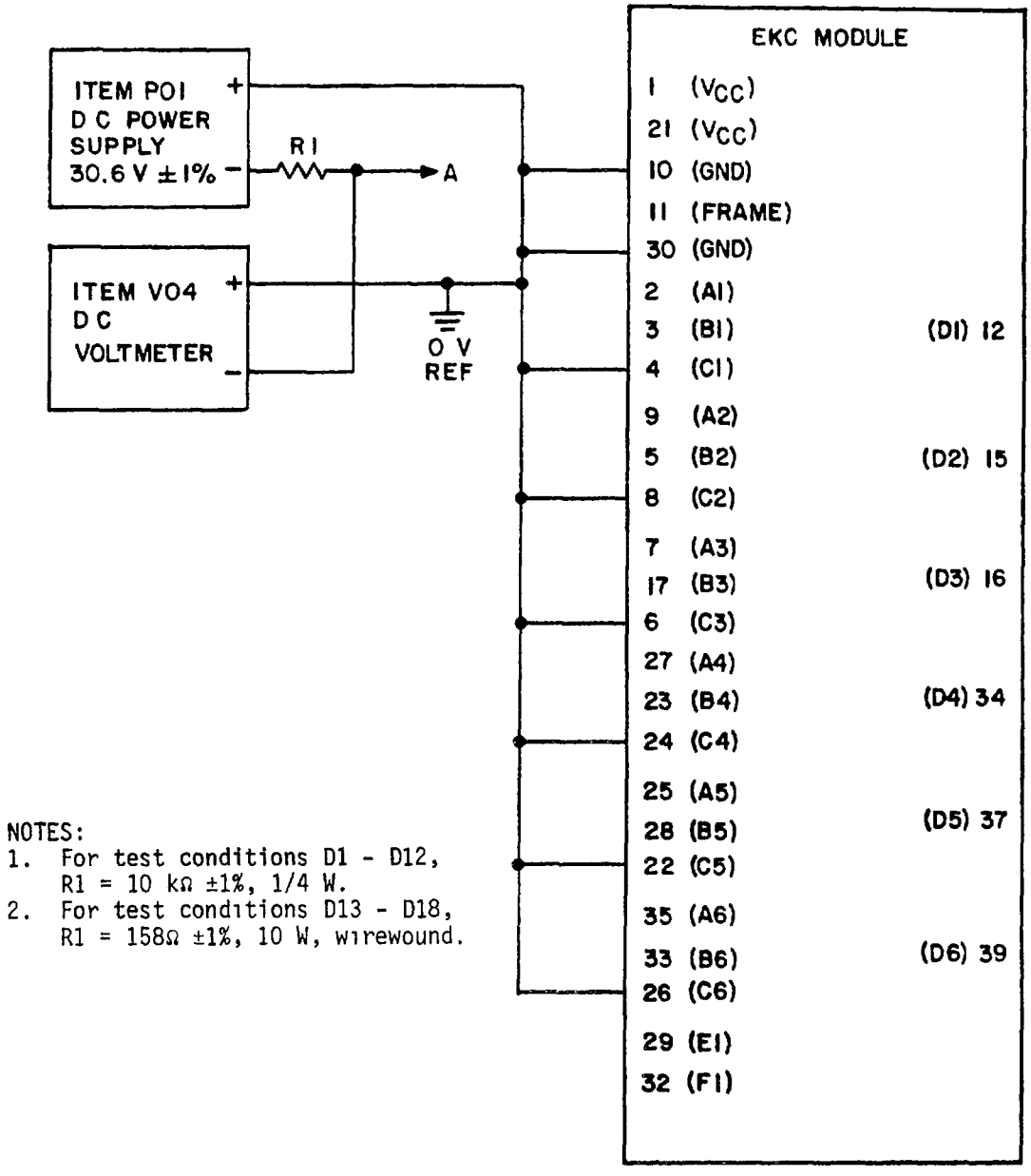
* FIGURE 9. Current limiting resistance and frame continuity test circuit for table V.

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*TABLE VI. Input and output protection diode voltage test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D1	B1/C	Input protection diode voltage (I_F = 3 mA)	3	500	710	mV	10
D2	B2/C	"	5	"	"	"	"
D3	B3/C	"	17	"	"	"	"
D4	B4/C	"	23	"	"	"	"
D5	B5/C	"	28	"	"	"	"
D6	B6/C	"	33	"	"	"	"
D7	D1	Output protection diode voltage (I_F = 3 mA)	12	"	750	"	"
D8	D2	"	15	"	"	"	"
D9	D3	"	16	"	"	"	"
D10	D4	"	34	"	"	"	"
D11	D5	"	37	"	"	"	"
D12	D6	"	39	"	"	"	"
D13	D1	Output protection diode voltage (I_F = 200 mA)	12	0.6	1.2	V	"
D14	D2	"	15	"	"	"	"
D15	D3	"	16	"	"	"	"
D16	D4	"	34	"	"	"	"
D17	D5	"	37	"	"	"	"
D18	D6	"	39	"	"	"	"

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NOTES:
1. For test conditions D1 - D12, R1 = 10 kΩ ±1%, 1/4 W.
2. For test conditions D13 - D18, R1 = 158Ω ±1%, 10 W, wirewound.

* FIGURE 10. Input and output protection diode voltage test circuit for table VI.

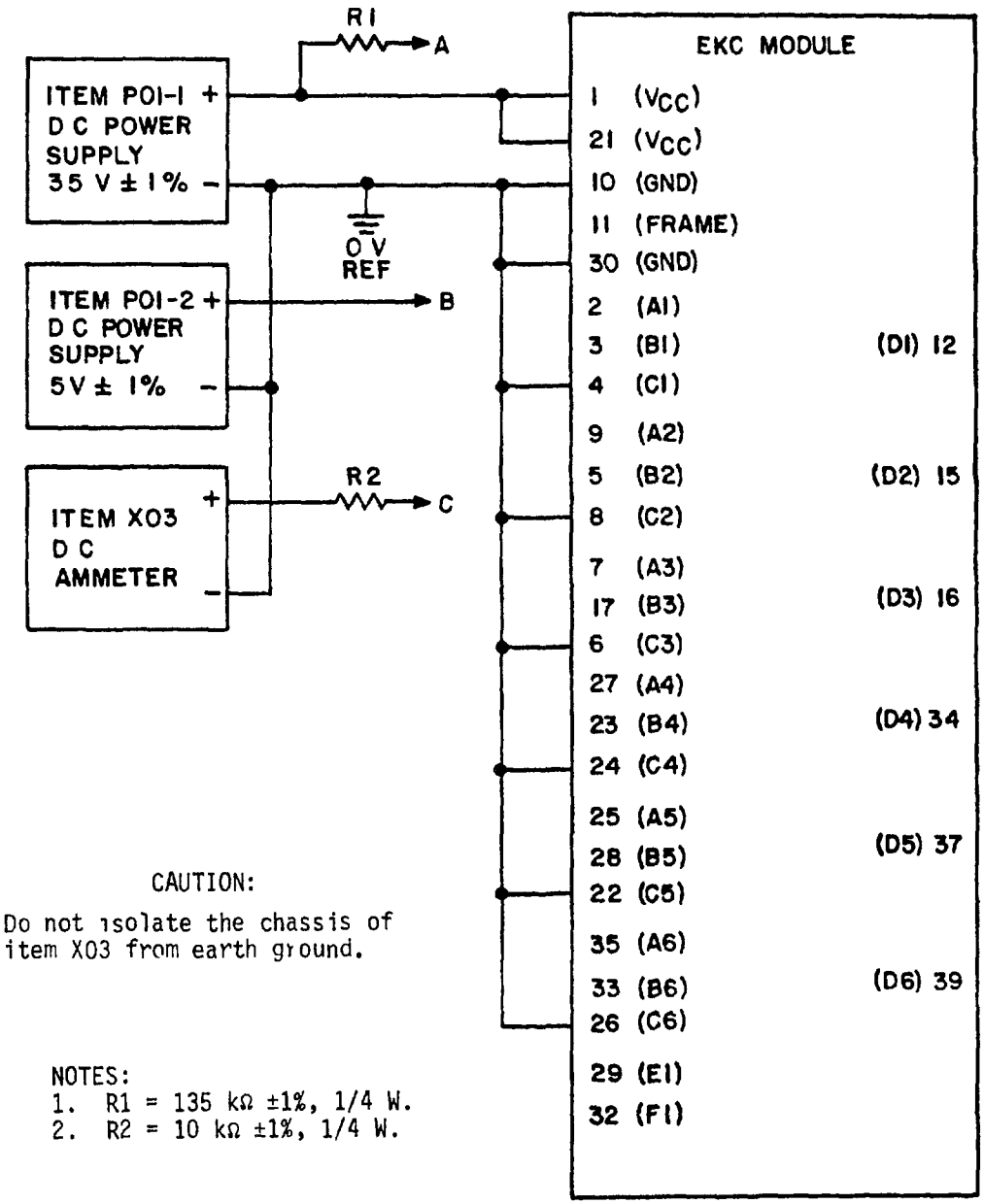
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TABLE VII. Output leakage current test. 1/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	D1 Output leakage current	2	9,7,27,25, 35	12	0	4	µA	11
E2	D2 "	9	2,7,27,25, 35	15	"	"	"	"
E3	D3 "	7	2,9,27,25, 35	16	"	"	"	"
E4	D4 "	27	2,9,7,25, 35	34	"	"	"	"
E5	D5 "	25	2,9,7,27, 35	37	"	"	"	"
E6	D6 "	35	2,9,7,27, 25	39	"	"	"	"

1/ These tests verify that with I_{IL} = 250 µA the output transistor is off ($I_{LEAKAGE}$ < 4 µA at V_{CC} = 35 V).

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CAUTION:

Do not isolate the chassis of item XO3 from earth ground.

NOTES:
1. R1 = 135 kΩ ±1%, 1/4 W.
2. R2 = 10 kΩ ±1%, 1/4 W.

FIGURE 11. Output leakage current test circuit for table VII.

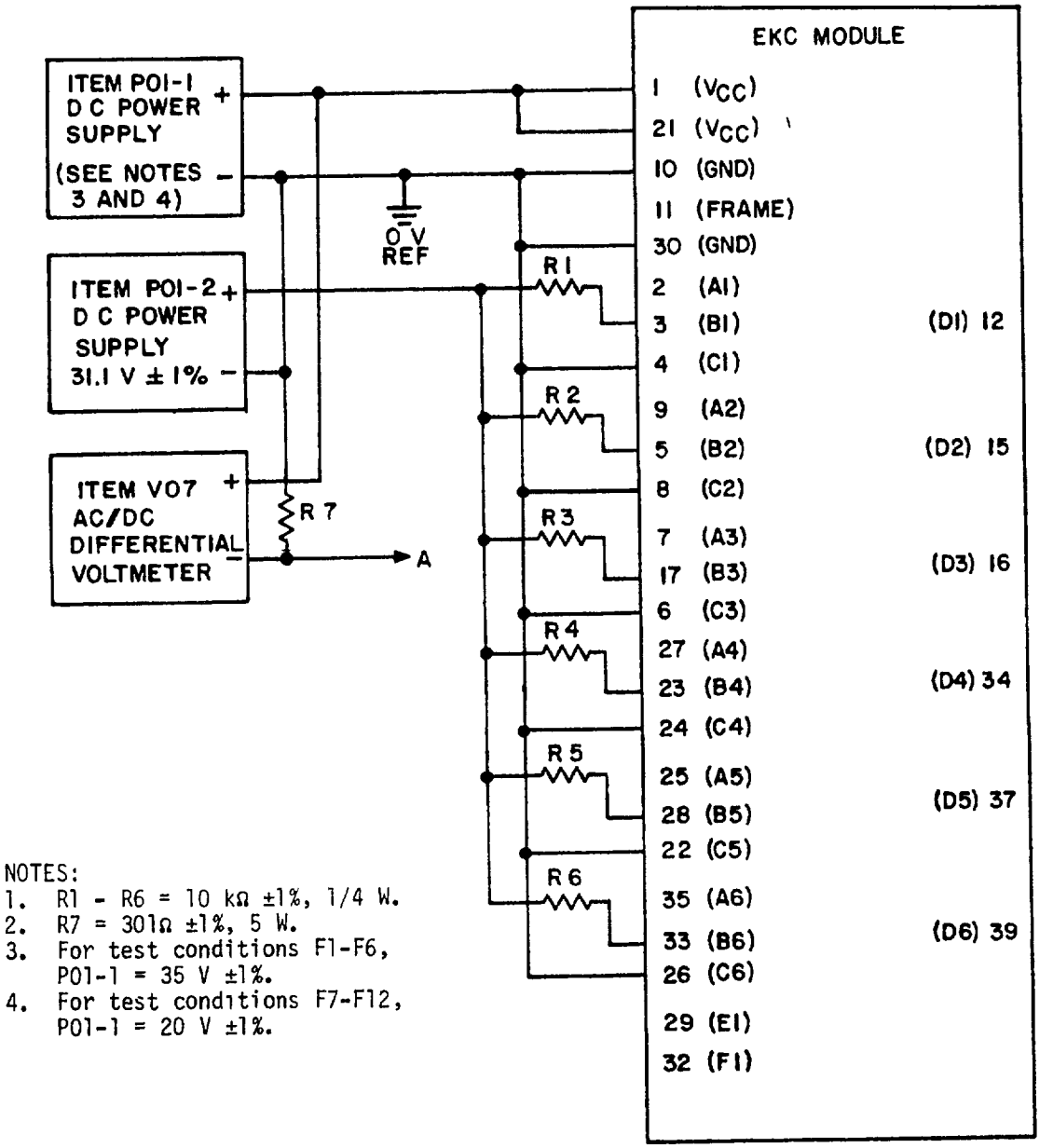
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*TABLE VIII. Output saturation voltage test. 1/

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	D1	Output saturation voltage	12	0	600	mV	12
F2	D2	"	15	"	"	"	"
F3	D3	"	16	"	"	"	"
F4	D4	"	34	"	"	"	"
F5	D5	"	37	"	"	"	"
F6	D6	"	39	"	"	"	"
F7	D1	"	12	"	500	"	"
F8	D2	"	15	"	"	"	"
F9	D3	"	16	"	"	"	"
F10	D4	"	34	"	"	"	"
F11	D5	"	37	"	"	"	"
F12	D6	"	39	"	"	"	"

1/ This test verifies that an I_{IH} = 3 mA turns on the output transistor ($V_{CE\ (sat)} \leq 600$ mV @ I_{OH} = 115 mA and $V_{CE\ (sat)} \leq 500$ mV @ I_{OH} = 65 mA).

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NOTES:
1. R1 - R6 = 10 kΩ ±1%, 1/4 W.
2. R7 = 301Ω ±1%, 5 W.
3. For test conditions F1-F6, P01-1 = 35 V ±1%.
4. For test conditions F7-F12, P01-1 = 20 V ±1%.

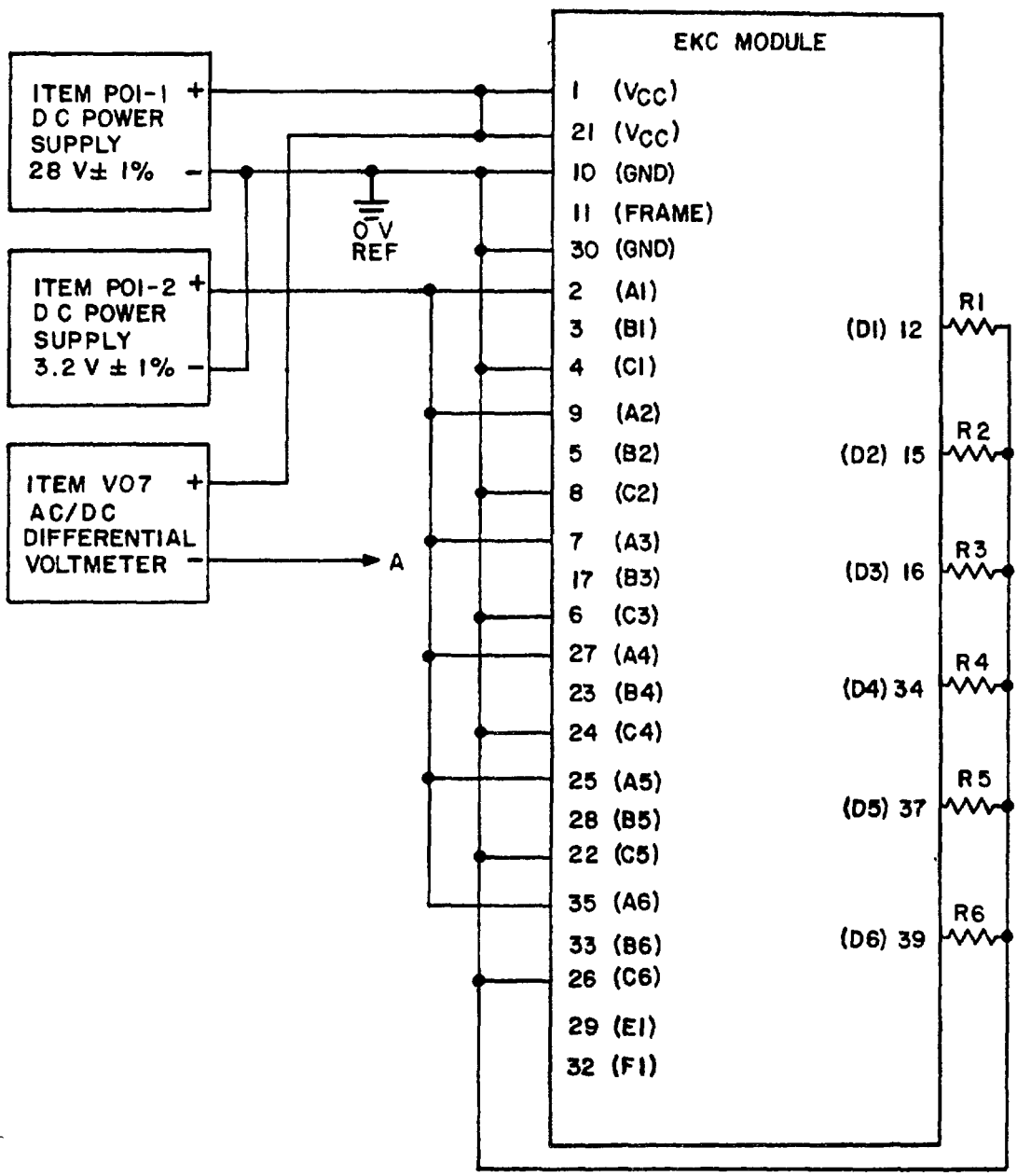
* FIGURE 12. Output saturation voltage test circuit for table VIII.

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TABLE IX. Output saturation voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G1	D1 Output saturation voltage (I_{OH} = 170 mA)	12	0	1	V	13
G2	D2 "	15	"	"	"	"
G3	D3 "	16	"	"	"	"
G4	D4 "	34	"	"	"	"
G5	D5 "	37	"	"	"	"
G6	D6 "	39	"	"	"	"

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NOTES:
1. R1 - R6 = 158Ω ±1%, 10 W, wirewound.
2. Item VO7 must be isolated from chassis ground.

FIGURE 13. Output saturation voltage test circuit for table IX.

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TABLE X. Power supply current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Power supply current, outputs on	1	D	E	17	25	mA	14
H2	Power supply current, outputs on	21	D	E	17	25	"	"
H3	Power supply current, outputs off	1	Open	Open	2.2	3.0	"	"
H4	Power supply current, outputs off	21	Open	Open	2.2	3.0	"	"

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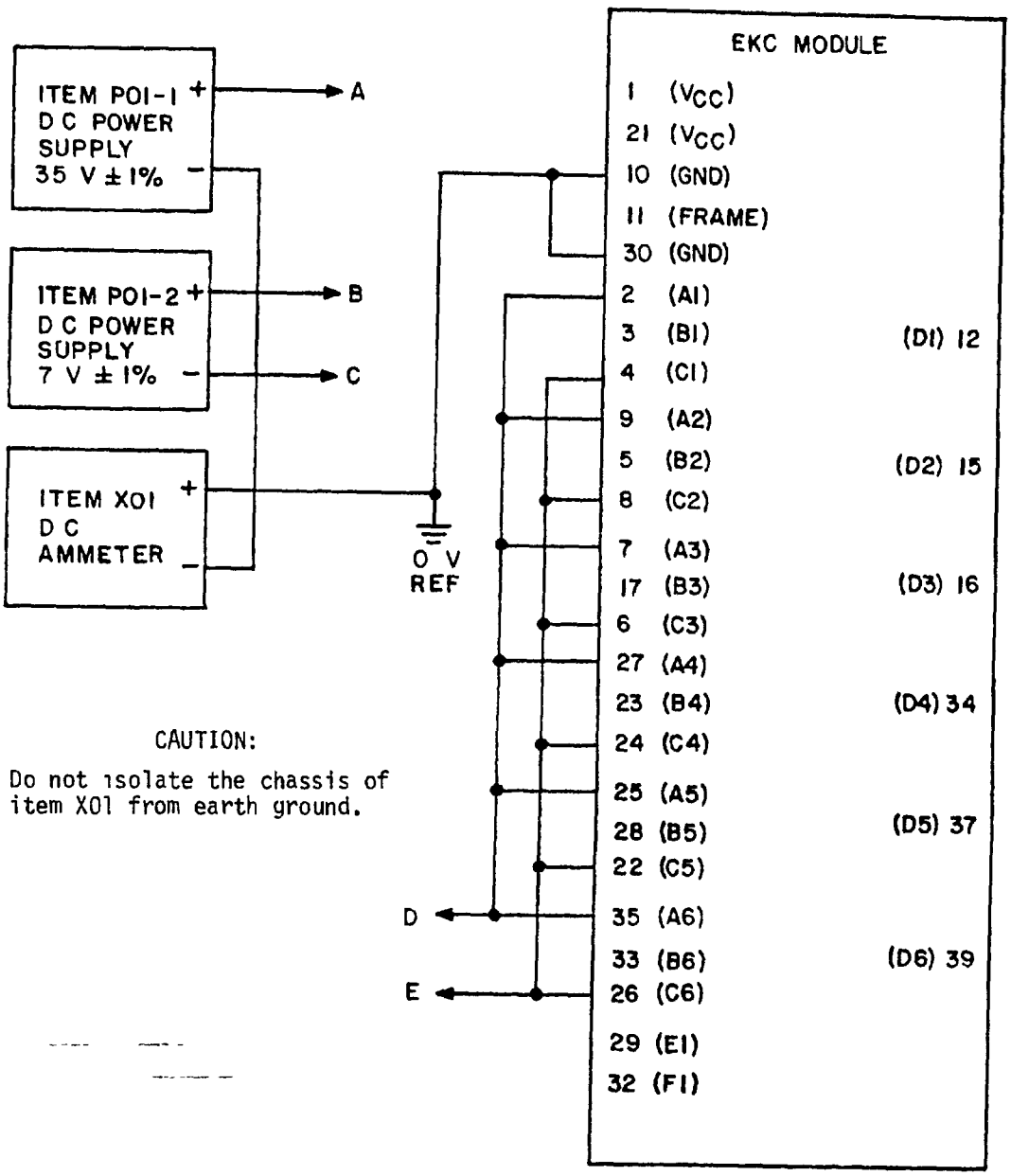


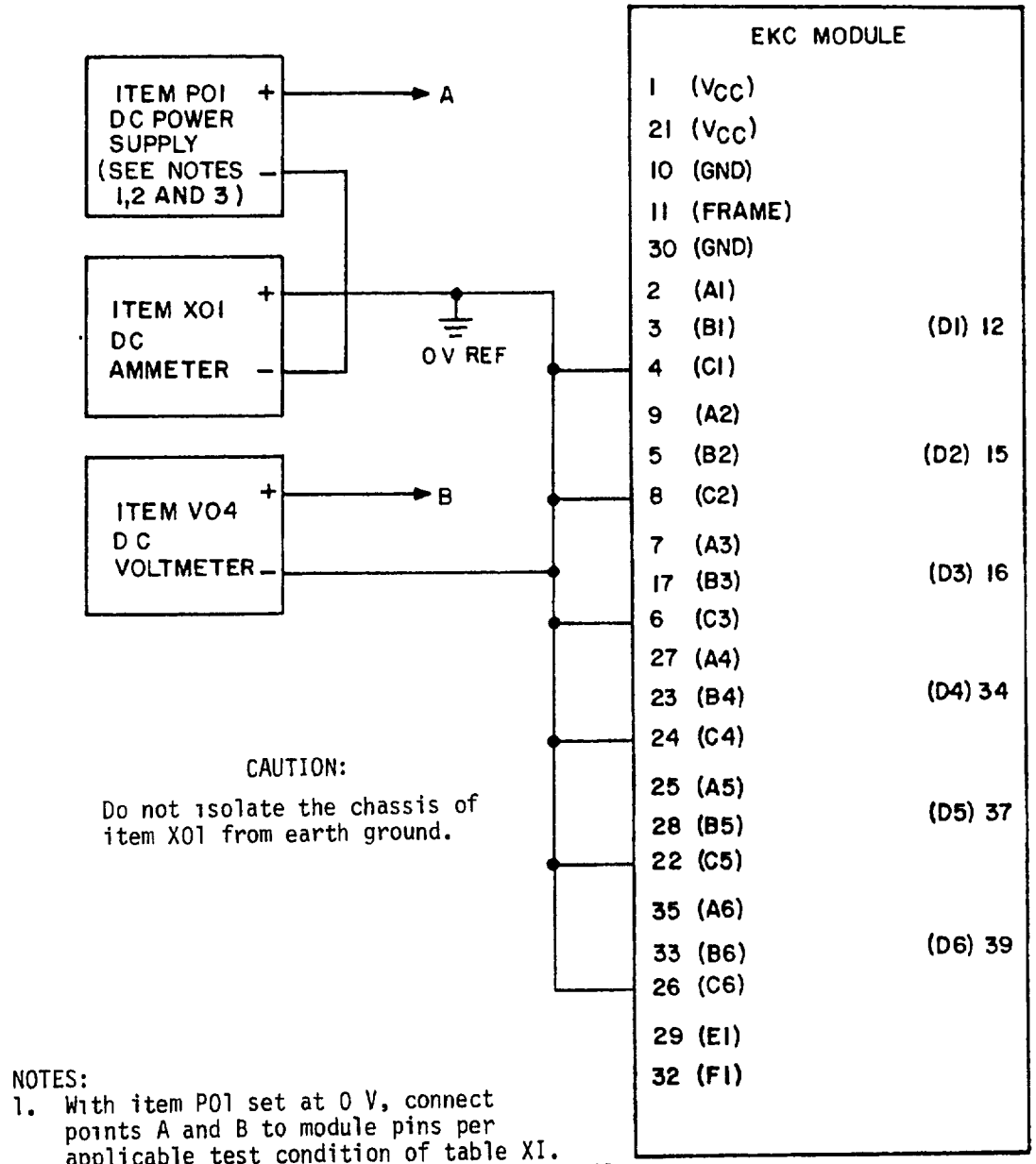
FIGURE 14. Power supply current test circuit for table X.

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TABLE XI. Input diode forward voltage test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	B1/C	Input diode forward voltage (I_F = 0.25 mA)	2	3	0.8	1.5	V	15
J2	B2/C	"	9	5	"	"	"	"
J3	B3/C	"	7	17	"	"	"	"
J4	B4/C	"	27	23	"	"	"	"
J5	B5/C	"	25	28	"	"	"	"
J6	B6/C	"	35	33	"	"	"	"
J7	B1/C	Input diode forward voltage (I_F = 10 mA)	2	3	"	"	"	"
J8	B2/C	"	9	5	"	"	"	"
J9	B3/C	"	7	17	"	"	"	"
J10	B4/C	"	27	23	"	"	"	"
J11	B5/C	"	25	28	"	"	"	"
J12	B6/C	"	35	33	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. With item P01 set at 0 V, connect points A and B to module pins per applicable test condition of table XI.
2. For test conditions J1-J6, adjust item P01 (not exceeding 2 V) for item X01 reading of 250 µA ±3%.
3. For test conditions J7-J12, adjust item P01 (not exceeding 10 V) for item X01 reading of 10 mA ±3%.

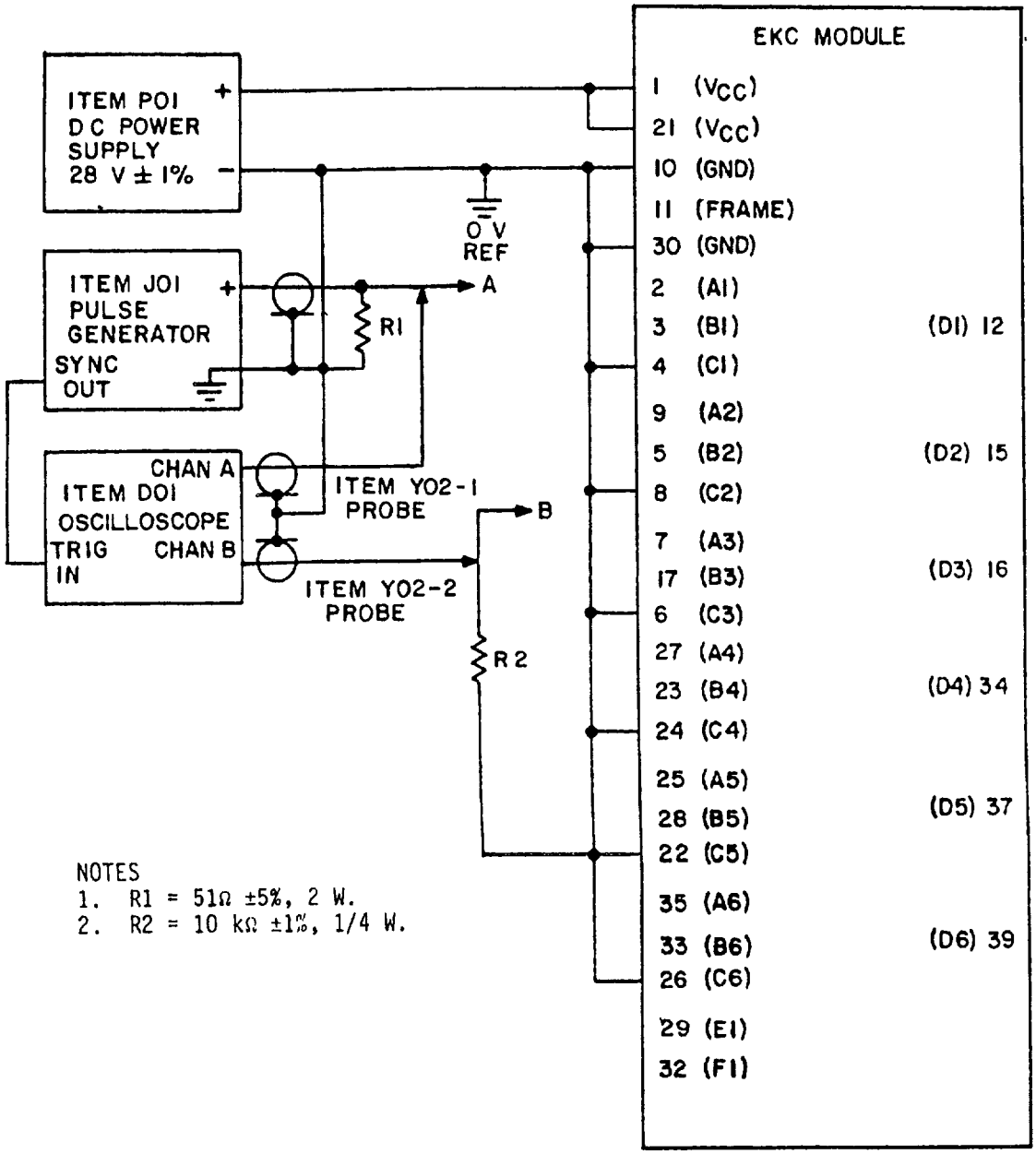
FIGURE 15. Input diode forward voltage test circuit for table XI.

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TABLE XII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	D1	Transition time (t_{TLH})	2	12	0.01	10.00	µs	16,17
K2	D2	"	9	15	"	"	"	"
K3	D3	"	7	16	"	"	"	"
K4	D4	"	27	34	"	"	"	"
K5	D5	"	25	37	"	"	"	"
K6	D6	"	35	39	"	"	"	"
L1	D1	Transition time (t_{THL})	2	12	0.1	100.0	"	"
L2	D2	"	9	15	"	"	"	"
L3	D3	"	7	16	"	"	"	"
L4	D4	"	27	34	"	"	"	"
L5	D5	"	25	37	"	"	"	"
L6	D6	"	35	39	"	"	"	"
M1	A1/C to D1	Propagation delay time (t_{PLH})	2	12	"	40.0	"	"
M2	A2/C to D2	"	9	15	"	"	"	"
M3	A3/C to D3	"	7	16	"	"	"	"
M4	A4/C to D4	"	27	34	"	"	"	"
M5	A5/C to D5	"	25	37	"	"	"	"
M6	A6/C to D6	"	35	39	"	"	"	"
N1	A1/C to D1	Propagation delay time (t_{PHL})	2	12	10	1,000	"	"
N2	A2/C to D2	"	9	15	"	"	"	"
N3	A3/C to D3	"	7	16	"	"	"	"
N4	A4/C to D4	"	27	34	"	"	"	"
N5	A5/C to D5	"	25	37	"	"	"	"
N6	A6/C to D6	"	35	39	"	"	"	"

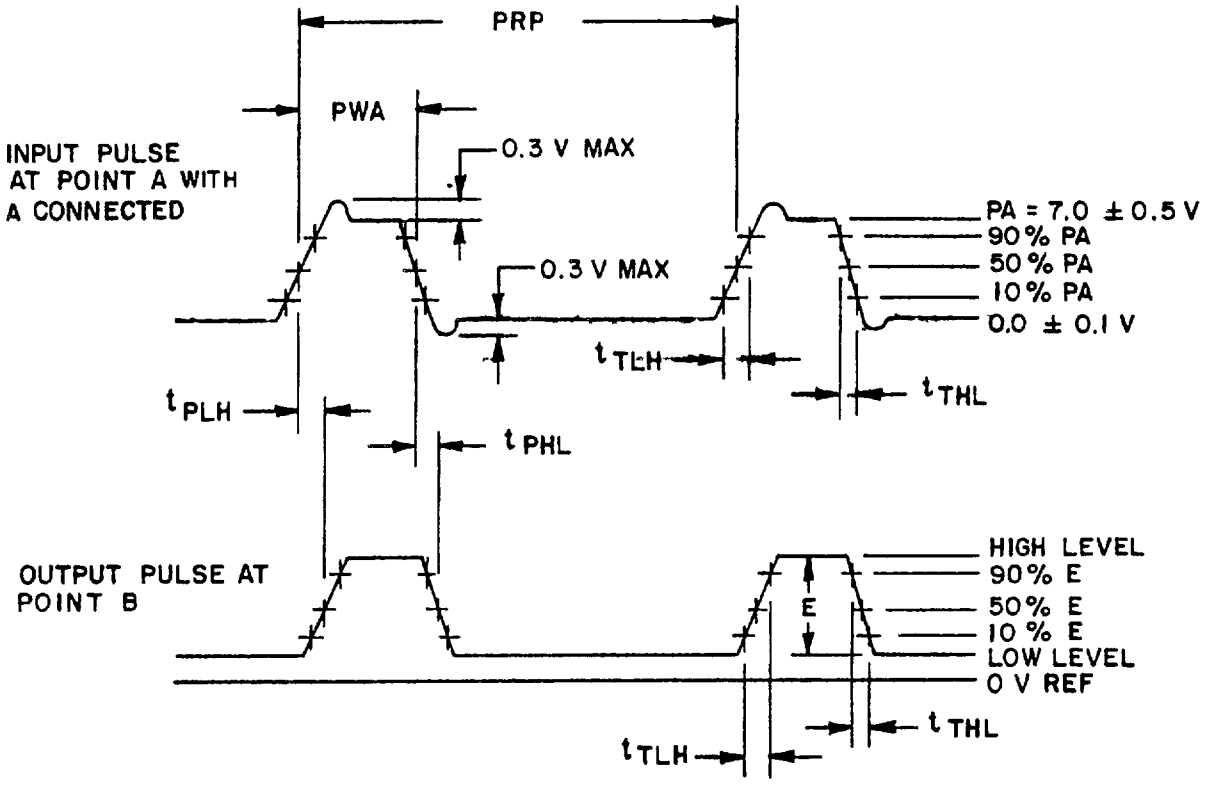
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NOTES
1. R1 = 51Ω ±5%, 2 W.
2. R2 = 10 kΩ ±1%, 1/4 W.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 25 \pm 5$ ns.
   PWA = 0.25 ±0.05 ms.
   PRP = 2.0 ±0.2 ms.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 27.4 V and the low level is $\leq$ 0.1 V.

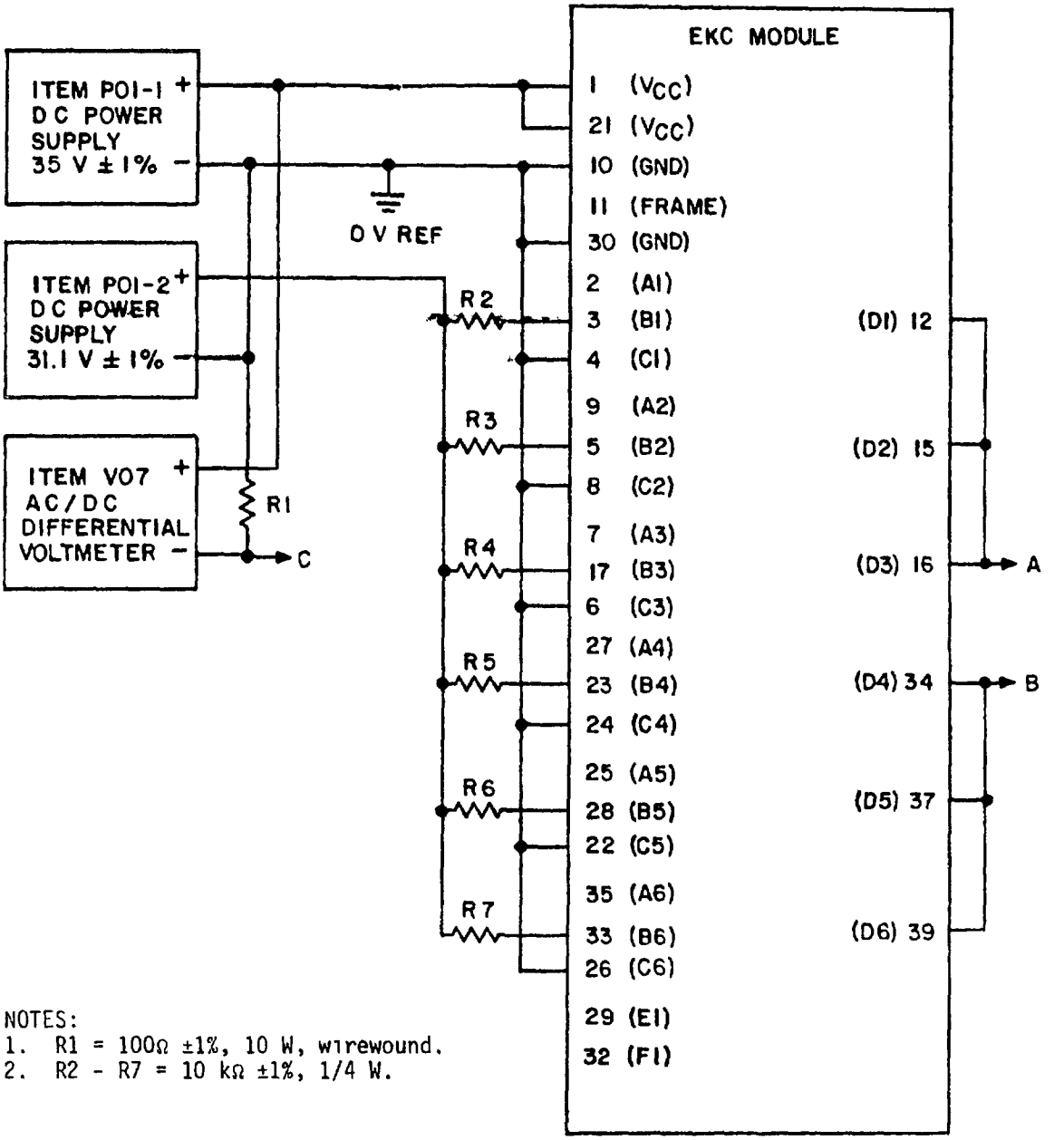
FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XII.

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*TABLE XIII. Parallel output saturation voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		C to point	Min	Max		
P1	Parallel output saturation voltage	A	0	600	mV	18
P2	Parallel output saturation voltage	B	0	600	mV	18

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NOTES:
1. R1 = 100Ω ±1%, 10 W, wirewound.
2. R2 - R7 = 10 kΩ ±1%, 1/4 W.

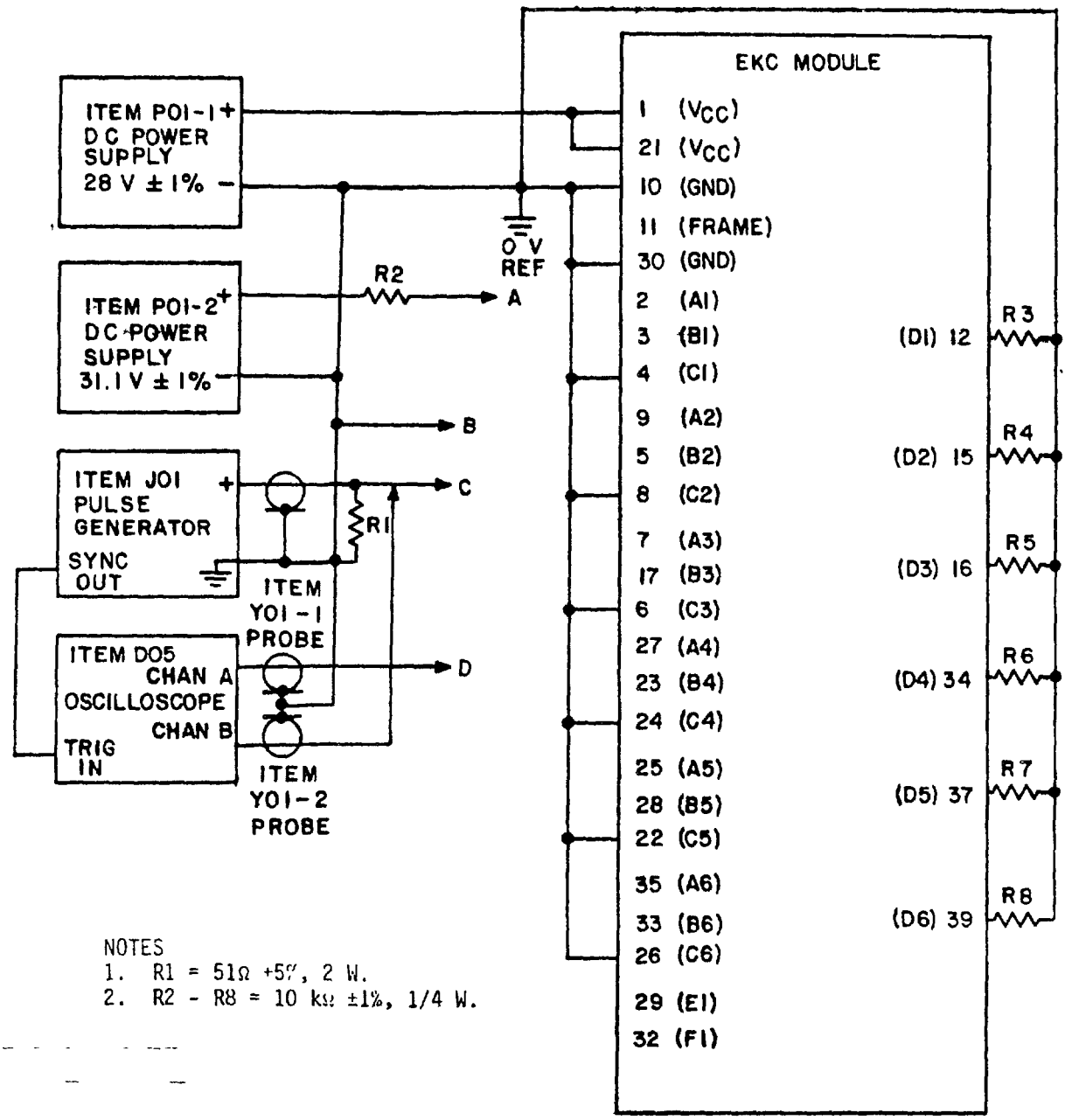
* FIGURE 18. Parallel output saturation voltage test circuit of table XIII.

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TABLE XIV. Output crosstalk (low-level and high-level) test.

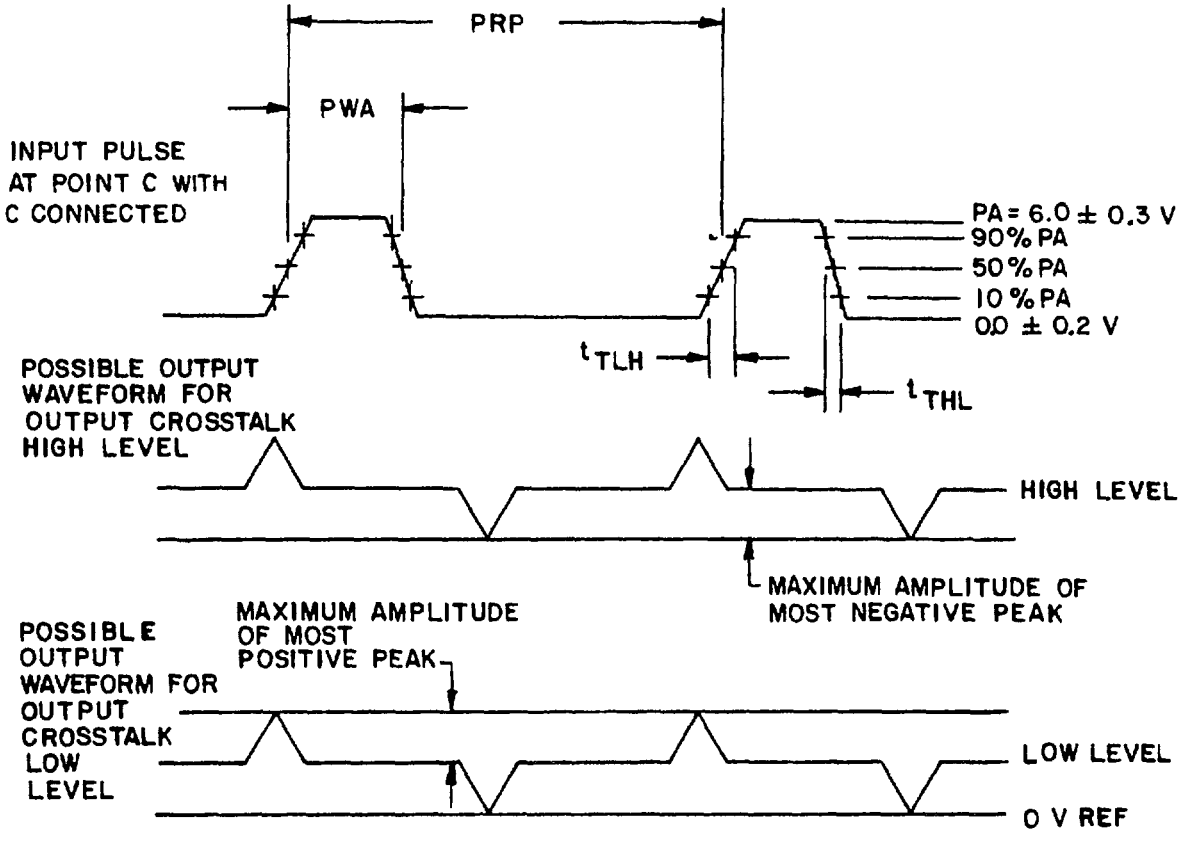
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	D1 Output crosstalk, low level	Open	3	7,9,25,27,35	12	0	2	V	19,20
Q2	D2 "	"	5	2,7,25,27,35	15	"	"	"	"
Q3	D3 "	"	17	2,9,25,27,35	16	"	"	"	"
Q4	D4 "	"	23	2,7,9,25,35	34	"	"	"	"
Q5	D5 "	"	28	2,7,9,27,35	37	"	"	"	"
Q6	D6 "	"	33	2,7,9,25,27	39	"	"	"	"
R1	D1 Output crosstalk, high level	3	Open	7,9,25,27,35	12	"	250	mV	"
R2	D2 "	5	"	2,7,25,27,35	15	"	"	"	"
R3	D3 "	17	"	2,9,25,27,35	16	"	"	"	"
R4	D4 "	23	"	2,7,9,25,35	34	"	"	"	"
R5	D5 "	28	"	2,7,9,27,35	37	"	"	"	"
R6	D6 "	33	"	2,7,9,25,27	39	"	"	"	"

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* FIGURE 19. Output crosstalk (low-level and high-level) test circuit for table XIV.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 5$ ns.
PWA = 0.5 ±0.1 ms.
PRP = 2.0 ±0.2 ms.

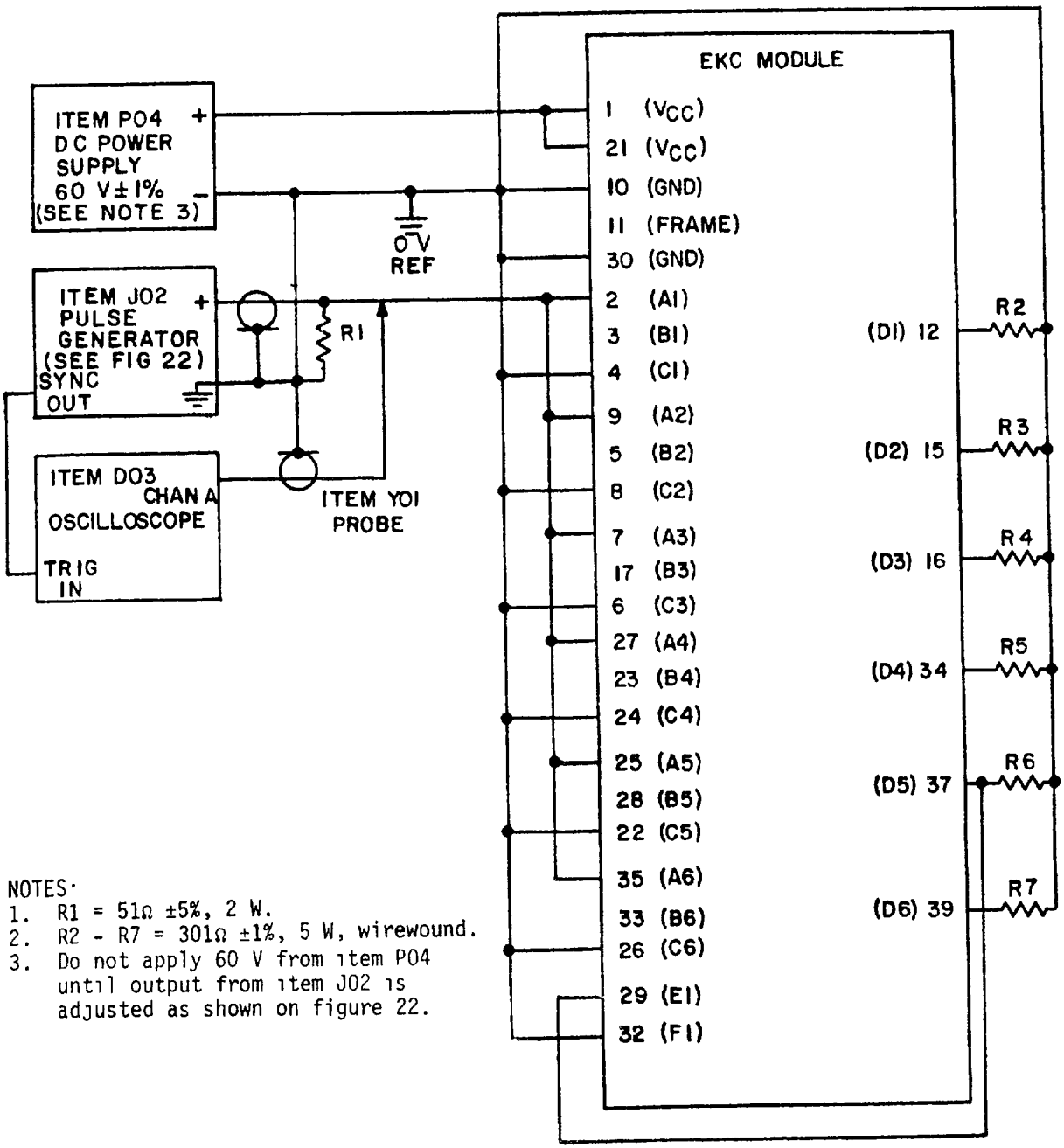
FIGURE 20. Output crosstalk (low-level and high-level) test circuit waveforms for table XIV.

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TABLE XV. Power supply short term overload test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Test figure
		Min	Max	
S	Power supply short term overload, V_{CC} = 60 V, R_L = 300 Ω	Must pass all functional tests - see 4.4.2		21,22

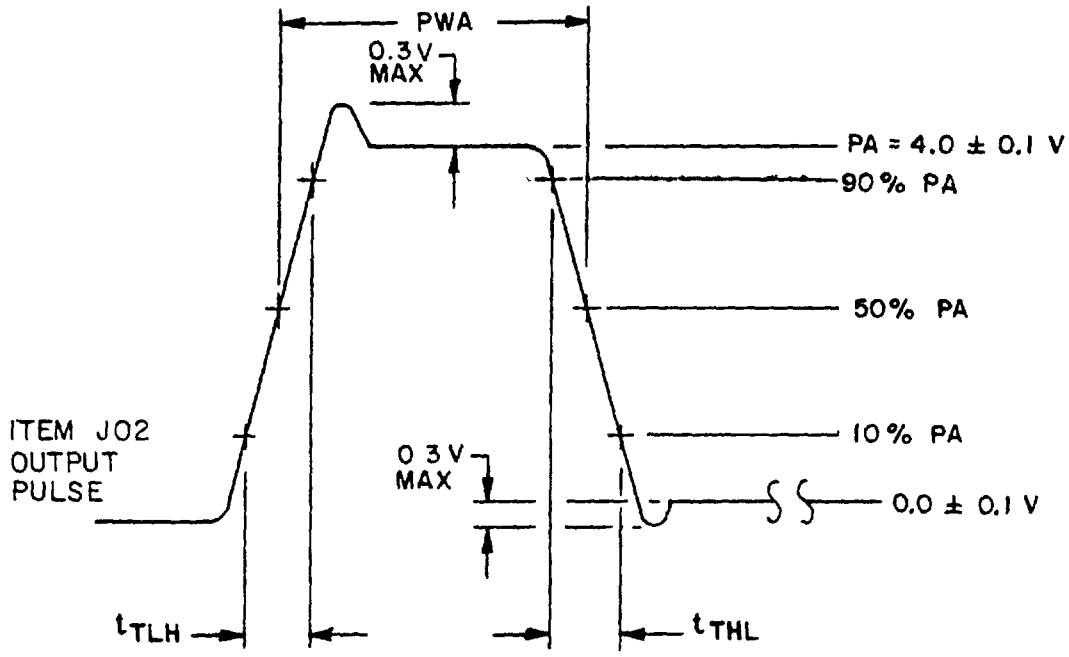
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NOTES:
1. R1 = 51Ω ±5%, 2 W.
2. R2 - R7 = 301Ω ±1%, 5 W, wirewound.
3. Do not apply 60 V from item P04 until output from item J02 is adjusted as shown on figure 22.

* FIGURE 21. Power supply short term overload test circuit for table XV.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 50 \pm 50$ µs.
PWA = 1.0 ±0.5 s.
PRP = One shot.

* FIGURE 22. Power supply short term overload test circuit waveform for table XV.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

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